



TELIPORT

Script Your Future

Season 4

Case Study No. 3

Team Details

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Team Name : bit_hackers

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THINK. BUILD. CREATE. WIN

IDEA SUBMISSION

Adaptive QoS Engine

PROBLEM

Mixed traffic such as video calls, gaming, downloads and browsing competes for bandwidth. Static allocation can increase congestion, latency and packet loss.

PROPOSED MODEL

An ML-assisted QoS engine classifies network traffic, predicts congestion risk and dynamically prioritizes bandwidth for latency-sensitive applications.

WORKFLOW



TECH STACK

Python · Scikit-learn · Pandas · NumPy
FastAPI / Streamlit · Network monitoring tools
ML classification · Congestion prediction · QoS policies

FUTURE SCOPE

Real-time deployment on routers and edge devices
Reinforcement learning for continuous QoS optimization
SDN integration, advanced analytics and network dashboards